Quantum Information

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Remark 1. *In this note all Hilbert spaces are complex and finite dimensional unless otherwise stated.*

1 Schmidt decomposition

Definition 1 (Schmidt decomposition)**.** *Let $\mathcal{H}, \mathcal{K}$ be Hilbert spaces, $|\Psi\rangle \in \mathcal{H} \otimes \mathcal{K}$. A Schmidt decomposition of $|\Psi\rangle$ is a finite set $\{(\lambda_i, |l_i\rangle, |r_i\rangle)\}_{i=1}^n \subseteq \mathbb{C} \times \mathcal{H} \times \mathcal{K}$ such that*

$$|\Psi\rangle = \sum_{i=1}^n \lambda_i \cdot |l_i\rangle \otimes |r_i\rangle,$$

and such that

- $\lambda_i > 0 \ \forall i = 1, \dots, n$,
- $\langle l_i | l_j \rangle = \delta_{ij}$ *for all* $i, j = 1, \dots, n$, *and*
- $\langle r_i | r_j \rangle = \delta_{ij}$ *for all* $i, j = 1, \dots, n$.

Theorem 1. *Let $\mathcal{H}, \mathcal{K}$ be Hilbert spaces, and $|\Psi\rangle \in \mathcal{H} \otimes \mathcal{K}$, $|\Psi\rangle \neq 0$. Then $|\Psi\rangle$ has a Schmidt decomposition.*

Proof. Let $\rho := \mathrm{Tr}_{\mathcal{K}} |\Psi\rangle\langle\Psi|$. This operator is positive semidefinite, and thus we can consider an eigen decomposition

$$\rho = \sum_i \lambda_i^2 \cdot |l_i\rangle\langle l_i|, \tag{1}$$

with $\lambda_i \in \mathbb{R}$, $\lambda_i \geq 0$ for all $i = 1 \dots \dim \mathcal{H}$. The vectors $\{|l_i\rangle\}_{i=1}^{\dim \mathcal{H}}$ form a basis of $\mathcal{H}$, we can thus write

$$|\Psi\rangle = \sum_i |l_i\rangle \otimes |\hat{r}_i\rangle,$$

for some $\{|\hat{r}_i\rangle\}_{i=1}^{\dim \mathcal{H}} \subseteq \mathcal{K}$. Expressing ρ with the help of this form of $|\Psi\rangle$, we obtain

$$\rho = \mathrm{Tr}_{\mathcal{K}} |\Psi\rangle\langle\Psi| = \sum_{i,j=1}^{\dim \mathcal{H}} |l_i\rangle\langle l_j| \cdot \mathrm{Tr}\{|\hat{r}_i\rangle\langle\hat{r}_j|\} = \sum_{i,j=1}^{\dim \mathcal{H}} |l_i\rangle\langle l_j| \cdot \langle\hat{r}_j|\hat{r}_i\rangle.$$

Comparing this to Eq. (1), we obtain $\langle\hat{r}_j|\hat{r}_i\rangle = \delta_{ij}\lambda_i^2$. For $\lambda_i \neq 0$ we can thus define $|r_i\rangle = \lambda_i^{-1}|\hat{r}_i\rangle$. These states are, by definition, orthonormal, and

$$|\Psi\rangle = \sum_{i:\lambda_i \neq 0} \lambda_i \cdot |l_i\rangle \otimes |r_i\rangle$$

is a Schmidt decomposition of $|\Psi\rangle$. $\square$

2 LOCC

We want to understand which states are more entangled than others. Our strategy: identify a sufficiently large class of operations (this will be LOCC) that we declare not to increase entanglement, then we say that $|\Psi\rangle$ is more entangled than $|\Phi\rangle$ iff we can transform $|\Psi\rangle$ to $|\Phi\rangle$ via those operations.

Which operations shall we choose? Quantum operations of (pure) states, thus they will consist of unitaries and measurements. Minimal requirement: do not transform separable states to entangled states. Unitaries are easy. Only product unitaries are allowed, as

Proposition 1. *Let $U \in \mathcal{B}(\mathcal{H} \otimes \mathcal{K})$. Then $U(|\psi\rangle \otimes |\phi\rangle)$ is a product state for all $|\psi\rangle \in \mathcal{H}$, $|\phi\rangle \in \mathcal{K}$, if and only if U is a product, i.e., there are $V \in \mathcal{B}(\mathcal{H})$ and $W \in \mathcal{B}(\mathcal{K})$ unitaries such that $U = V \otimes W$.*

Proof. Exercise. □

Theorem 2 (Ky-Fan)**.** *Let $\mathcal{H}$ be a Hilbert space, $k \in \mathbb{N}$, and*

$$\mathcal{P}_k(\mathcal{H}) := \left\{ P \in \mathcal{B}(\mathcal{H}) \middle| P = P^\dagger = P^2,\ \mathrm{Tr}(P) = k \right\}.$$

Let $\rho \in \mathcal{B}(\mathcal{H})$ Hermitian, and

$$\rho = \sum_{i=1}^{\dim \mathcal{H}} \lambda_i |\Phi_i\rangle\langle\Phi_i|$$

be its eigen decomposition, with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{\dim \mathcal{H}}$. Then

$$\sup_{P \in \mathcal{P}_k} \mathrm{Tr}\{\rho P\} = \sum_{i=1}^{k} \lambda_i.$$

In other words, given a Hermitian operator ρ, the maximum of $\mathrm{Tr}\{\rho P\}$, where P is a rank-k Hermitian projector, is the sum of the k largest eigenvalues of ρ.

Proof. Let $Q = \sum_i |\Phi_i\rangle\langle\Phi_i|$. Then $Q \in \mathcal{P}_k$ and thus

$$\sup_{P \in \mathcal{P}_k} \mathrm{Tr}\{\rho P\} \geq \mathrm{Tr}\{Q\rho\} = \sum_{i=1}^{k} \langle\Phi_i|\rho|\Phi_i\rangle = \sum_{i=1}^{k} \lambda_i. \tag{2}$$

To obtain the opposite bound, use the spectral decomposition of ρ to write

$$\mathrm{Tr}\{\rho P\} = \sum_{i=1}^{\dim \mathcal{H}} \lambda_i \,\mathrm{Tr}\left\{P|\Phi_i\rangle\langle\Phi_i|\right\} = \sum_{i=1}^{\dim \mathcal{H}} \lambda_i \cdot \langle\Phi_i|P|\Phi_i\rangle = \sum_{i=1}^{\dim \mathcal{H}} \lambda_i \cdot \omega_i,$$

where $\omega_i = \langle\Phi_i|P|\Phi_i\rangle$ satisfies $0 \leq \omega_i \leq 1$ and $\sum_{i=1}^{k} \omega_i = k$. We thus obtain that

$$\sup_{P \in \mathcal{P}_k} \mathrm{Tr}\{\rho P\} \leq \sup_{\substack{\omega \in [0,1]^{\dim \mathcal{H}} \\ \sum_i \omega_i = k}} \sum_{i=1}^{\dim \mathcal{H}} \lambda_i \cdot \omega_i.$$

The maximum of the r.h.s. is reached at $\omega_1 = \cdots = \omega_k = 1$, and $\omega_{k+1} = \cdots = \omega_{\dim \mathcal{H}} = 0$, and its value is $\sum_{i=1}^{k} \lambda_i$. Therefore

$$\sup_{P \in \mathcal{P}_k} \mathrm{Tr}\{\rho P\} \leq \sup_{\substack{\omega \in [0,1]^{\dim \mathcal{H}} \\ \sum_i \omega_i = k}} \sum_{i=1}^{\dim \mathcal{H}} \lambda_i \cdot \omega_i = \sum_{i=1}^{k} \lambda_i.$$

This, together with Eq. (2) is the desired statement. □

Definition 2 (Majorization)**.** *Let $p, q \in \mathbb{R}^n$ be probability distributions. Let $p^\downarrow$ ($q^\downarrow$) be the vector obtained by listing the entries of p (q) in descending order. We say that p is majorized by q, and write $p \preccurlyeq q$, if for all $k = 1, \ldots, n$,*

$$\sum_{i=1}^{k} p_i^\downarrow \leq \sum_{i=1}^{k} q_i^\downarrow.$$

TODO: Here it is better to define Schmidt coefficient with zero in it.

Proposition 2. *Let $|\Psi\rangle, |\Phi\rangle \in \mathcal{H} \otimes \mathcal{K}$. Let the squared Schmidt coefficients of $|\Psi\rangle$ be p, and the squared Schmidt coefficients of $|\Phi\rangle$ be q. If $|\Psi\rangle \xrightarrow{LOCC} |\Psi\rangle$, then $q \preccurlyeq p$.*

Proof. Let $\rho = \mathrm{Tr}_{\mathcal{H}} |\Psi\rangle\langle\Psi|$ and $\eta = \mathrm{Tr}_{\mathcal{H}} |\Psi\rangle\langle\Psi|$. Then the eigenvalues of ρ are precisely the entries of p and the eigenvalues of η are the entries of q. Using MISSING, $|\Psi\rangle \xrightarrow{LOCC} |\Psi\rangle$ if and only if

$$\sqrt{r_i}|\Phi\rangle = M_i \otimes U_i|\Psi\rangle,$$

for a measurement $\{M_i\}_{i=1}^n$, a probability distribution $r \in \mathbb{R}^n$ and some unitaries $\{U_i\}_{i=1}^n$. Therefore

$$r_i \eta = U_i \rho_i U_i^\dagger,$$

where $\rho_i = \mathrm{Tr}_{\mathcal{H}}\{(M_i^\dagger M_i \otimes \mathrm{Id})|\Psi\rangle\langle\Psi|\}$, and thus

$$\sup_{P \in \mathcal{P}_k} \mathrm{Tr}\{P\eta\} = \sup_{P \in \mathcal{P}_k} \sum_i \mathrm{Tr}\{r_i P \eta\} = \sup_{P \in \mathcal{P}_k} \sum_i \mathrm{Tr}\{P U_i \rho_i U_i^\dagger\} \leq \sum_i \sup_{P \in \mathcal{P}_k} \mathrm{Tr}\{U_i^\dagger P U_i \rho\} = \sum_i \sup_{P \in \mathcal{P}_k} \mathrm{Tr}\{P \rho_i\} = \sup_{P \in \mathcal{P}_k} \mathrm{Tr}\{P\rho\}.$$

Using now Theorem 2, we obtain that $p \preccurlyeq q$. □

Let us try to understand now a *sufficient* criterion for $|\Psi\rangle \xrightarrow{LOCC} |\Phi\rangle$.

Example 1. Let $|\Psi\rangle = \sqrt{1/2}|00\rangle + \sqrt{1/2}|11\rangle$ and $|\Phi\rangle = \sqrt{2/3}|00\rangle + \sqrt{1/3}|11\rangle$. Notice that

$$M_0 = \begin{pmatrix} \sqrt{2/3} & 0 \\ 0 & \sqrt{1/3} \end{pmatrix},$$

is a matrix such that $(M_0 \otimes \mathrm{Id})|\Psi\rangle = |\Phi\rangle$. Moreover, we can complete it to a measurement, for example, with the matrix

$$M_1 = \begin{pmatrix} \sqrt{1/3} & 0 \\ 0 & \sqrt{2/3} \end{pmatrix},$$

i.e., $M_0^\dagger M_0 + M_1^\dagger M_1 = \mathrm{Id}$. Notice that $(M_1 \otimes \mathrm{Id})|\Psi\rangle = X \otimes X|\Phi\rangle$. Therefore the measurement $N_i = X^i M_i$ ($i = 1, 2$) is such that

$$M_i \otimes X^i|\Psi\rangle = \sqrt{p_i}|\Phi\rangle,$$

for some probability distribution p_i, i.e., $|\Psi\rangle \xrightarrow{LOCC} |\Phi\rangle$.

The essence of this example is that the squared Schmidt coefficients of the states are related to each other as

$$\begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix} = \frac{1}{2} \cdot (\mathrm{Id} + X) \cdot \begin{pmatrix} 1/3 \\ 2/3 \end{pmatrix}.$$

This construction can be easily generalized for the case when the squared Schmidt coefficients of $|\Psi\rangle$ can be obtained by mixing permutations of the squared Schmidt coefficients of $|\Phi\rangle$. Before we show how to obtain the LOCC operations for this generalized case, notice that this constraint is equivalent with majorization:

Lemma 1. *Let $p, q \in \mathbb{R}^n$ be probability distributions. If $p \preccurlyeq q$, then there is a probability distribution $r \in \mathbb{R}^{n!}$ such that*

$$p = \sum_{\pi \in S_n} r_\pi \pi q,$$

where $S_n \subseteq \mathcal{B}(\mathbb{R}^n)$ is the set of $n \times n$ permutation matrices.

Proof. Exercise. □

Again, number of non-zero Schmidt coeffs might be different!

Proposition 3. *Let $|\Psi\rangle, |\Phi\rangle \in \mathcal{H} \otimes \mathcal{K}$. Let the squared Schmidt coefficients of $|\Psi\rangle$ be p, and the squared Schmidt coefficients of $|\Phi\rangle$ be q. If $p \preccurlyeq q$, then $|\Psi\rangle \xrightarrow{LOCC} |\Phi\rangle$.*

Proof. Using Lemma 1, we obtain that there is a probability distribution r such that $p = \sum_{\pi \in S_n} r_\pi \pi q$, where S_n is the group of $n \times n$ permutation matrices. Let the Schmidt decomposition of the states be

$$|\Psi\rangle = \sum_{i=1}^{n} p_i \cdot |l_i\rangle \otimes |r_i\rangle$$
$$|\Phi\rangle = \sum_{i=1}^{n} q_i \cdot |\hat{l}_i\rangle \otimes |\hat{r}_i\rangle,$$

and let us define operators M_π as

$$M_\pi = \sum_i \left(r_\pi \frac{q_{\pi(i)}}{p_i} \right)^{1/2} |\hat{l}_{\pi(i)}\rangle\langle l_i|.$$

These operators form a POVM as

$$\sum_\pi M_\pi^\dagger M_\pi = \sum_{ij} \sum_\pi r_\pi \frac{q_{\pi(i)}}{p_i} |l_i\rangle\langle \hat{l}_{\pi(i)} | \hat{l}_{\pi(j)}\rangle\langle l_j| = \sum_i \sum_\pi r_\pi \frac{q_{\pi(i)}}{p_i} |l_i\rangle\langle l_i| = \sum_i |l_i\rangle\langle l_i| = \mathrm{Id}.$$

Note as well that

$$(M_\pi \otimes \mathrm{Id})|\Psi\rangle = \sum_i \sqrt{r_\pi q_{\pi(i)}} |\hat{l}_{\pi(i)}\rangle \otimes |r_i\rangle,$$

and thus, setting $V_\pi = \sum_i |r_{\pi(i)}\rangle\langle r_i|$, V_π is a unitary and

$$(M_\pi \otimes V_\pi)|\Psi\rangle = \sum_i \sqrt{r_\pi q_{\pi(i)}} |\hat{l}_{\pi(i)}\rangle \otimes |r_{\pi(i)}\rangle = \sqrt{r_\pi} |\Phi\rangle,$$

i.e., we have shown that $|\Psi\rangle \xrightarrow{LOCC} |\Phi\rangle$. □

We have thus seen in Proposition 2 and Proposition 3 that

Theorem 3. *Let $|\Psi\rangle, |\Phi\rangle \in \mathcal{H} \otimes \mathcal{K}$. Let the squared Schmidt coefficients of $|\Psi\rangle$ be p, and the squared Schmidt coefficients of $|\Phi\rangle$ be q. Then $|\Psi\rangle \xrightarrow{LOCC} |\Phi\rangle$ iff $p \preccurlyeq q$.*